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Problem with Sensitive Data

- Handling sensitive data securely is challenging
- External APIs may expose confidential information
- Compliance and data protection requirements (e.g. GDPR)

Solution: Self Hosting

- Full control over data and infrastructure
- No data leaves the organization
- Customizable and scalable setup

Self Hosting: Architecture Overview

- Multiple services working together
- Containerized and orchestrated infrastructure
- Focus on performance, privacy, and extensibility

Self Hosting: NGINX

- Serves static landing page
- Acts as reverse proxy to route traffic
- Maps subdomains to organization-specific services

Self Hosting: Open WebUI

- Acts as frontend
- Users can chat with their data (RAG)
- Trigger N8N workflows via tools

Self Hosting: vLLM

- Runs chat model (e.g., GPT-OSS 20B) and reranker
- OpenAI-compatible API
- High-performance:
 - PagedAttention
 - Continuous batching
 - Quantization
- Supports models from Hugging Face

Self Hosting: PostgreSQL

- Persistent storage for:
 - User data (email, password)
 - Chat history
 - API keys
 - Embeddings for RAG

Self Hosting: n8n

- No-code / low-code workflow automation
- Triggered via Open WebUI webhooks
- Integrates with third-party tools:
 - WhatsApp
 - Telegram
 - Microsoft Teams
 - Outlook

Self Hosting: Docker

- Packages services into containers
- Ensures portability
- Simplifies deployment
- Managed via `docker-compose.yaml`

Self Hosting: Kubernetes

- Orchestrates containers (pods)
- Automatically restarts failed services
- Enables zero-downtime updates

Requirements / Demo

- Infrastructure setup
- GPU resources for LLMs
- Secure networking
- Live demonstration

Thank You!